

Bridging characters: betweenness centrality in co-occurrence networks of classic English novels

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Abstract

Networks extracted from works of fiction provide a quantitative window into narrative structure. Here we model ten canonical English novels from Project Gutenberg as weighted character co-occurrence networks, where two characters are linked whenever they are mentioned within a sliding window of eight paragraphs. The main methodological difficulty is not the network analysis itself but the reliable identification of characters: we rely on a neural literary natural-language-processing pipeline (BookNLP) that performs named-entity recognition, alias resolution (e.g. *Elizabeth*, *Lizzy* and *Miss Bennet* collapse into a single vertex) and coreference resolution, so that anaphoric pronouns contribute to the presence of the correct character. Across all books the resulting networks display a pronounced small-world signature, with high clustering ($C \approx 0.69$ – 0.79), short average path lengths ($L \approx 1.6$ – 2.1) and small-world coefficients σ between 2 and 9.5, together with disassortative degree mixing and clear community structure. We rank characters by betweenness centrality and show that the most “between” character is the narrative connector rather than simply the most frequent one: in first-person novels, where the narrator is realised mostly through pronouns, a secondary character (e.g. *Jim* in *Huckleberry Finn* or *Joe* in *Great Expectations*) emerges as the structural bridge. Betweenness is also the centrality least correlated with the others, confirming that it captures a distinct, mediation-oriented notion of importance.

Keywords: complex networks, character networks, betweenness centrality, natural language processing, small-world, digital humanities

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1. Introduction

The application of network science to written texts has become a productive research line, ranging from word-adjacency networks for stylometry and authorship attribution to the identification of literary movements [1, 2, 3]. A complementary representation focuses not on words but on the *entities* that populate a text: by linking named entities that appear in the same context, one obtains a co-occurrence network whose topology carries information that is largely orthogonal to word-level descriptors [4]. When the entities are the characters of a work of fiction, such networks become *character networks*, a representation that has been used for tasks as diverse as summarisation, classification and role detection [5, 6].

A character network is a graph whose vertices are the characters of a narrative and whose edges encode some notion of interaction or proximity between them [5]. Once the network is built, the toolbox of complex networks can be brought to bear: degree, clustering, community structure and, crucially, centrality measures that quantify the structural importance of each character. Among these, *betweenness centrality* [7, 8] is particularly appealing for narrative analysis because it measures the extent to which a vertex lies on the shortest paths between other vertices, i.e. the degree to which a character *bridges* otherwise weakly connected parts of the story.

In this work we build weighted character co-occurrence networks for ten classic English novels and ask a deceptively simple question: *which character has the highest betweenness centrality in each book?* Answering it reliably requires solving several natural-language-processing problems first — recognising which entities are characters, merging the many aliases of a single character, and resolving anaphoric pronouns to their referents. We address these with BookNLP, a pipeline specifically designed for English fiction whose entity and coreference components build on the literary-character model of Bamman et al. [9]. Our contributions are: (i) a fully reproducible pipeline from raw Gutenberg text to analysed network; (ii) a characterisation of the global topology of the ten networks, which turn out to be markedly small-world; and (iii) an analysis of betweenness that reveals the structural role of *connector* characters and its interaction with narrative point of view.

2. Related work

The extraction and analysis of fictional character networks is reviewed comprehensively by Labatut and Bost [5], who distinguish *conversational*, *co-occurrence* and *affiliation* networks according to how edges are defined. Early influential work by Elson et al. [6] extracted conversational networks from nineteenth-century British fiction by attributing quoted speech to speakers. Our edges instead follow the co-occurrence convention: two characters are linked when they are mentioned close together in the text, a criterion that is robust to the difficulties of speech attribution and aligns with the named-entity co-occurrence networks of Amancio [4], who showed that such networks exhibit a small-world structure and provide complementary information for text characterisation.

The reliability of any character network hinges on the quality of the underlying entity processing. We use BookNLP, whose statistical character model originates in the Bayesian mixed-effects formulation of literary character by Bamman et al. [9], and which internally relies on modern NLP components in the spirit of [10]. For the network side we adopt standard measures and algorithms [7, 8, 11] and interpret the global topology through the small-world lens of Watts and Strogatz [12].

3. Materials and methods

3.1. Corpus

We selected ten widely studied novels in English, downloaded as plain UTF-8 text from Project Gutenberg [13]. The Gutenberg header and licence footer were stripped, keeping only the body of each work. Table 1 summarises the corpus, including the number of tokens and paragraphs after cleaning.

Table 1: The ten-novel corpus and the size of the extracted networks.

Novel	Author	Year	Tokens	Paragraphs	Characters	Edges
Pride and Prejudice	Austen	1813	152,543	2,509	94	1741
Moby-Dick	Melville	1851	260,546	2,802	245	2716
Frankenstein	Shelley	1818	85,881	795	50	297
A Tale of Two Cities	Dickens	1859	168,428	3,327	91	769
Dracula	Stoker	1897	191,530	2,190	112	1019
Great Expectations	Dickens	1861	226,689	3,915	121	1182
Huckleberry Finn	Twain	1884	140,722	2,275	124	893
The Adventures of Sherlock Holmes	Doyle	1892	127,667	2,546	105	531
Emma	Austen	1815	189,463	2,382	120	2031
Jane Eyre	Brontë	1847	230,455	4,115	147	1420

3.2. Character identification, alias and anaphora resolution

The central NLP challenge is to turn raw prose into a clean set of characters. We process each novel with BookNLP (English, `small` model), which performs, in a single pass: (i) named-entity recognition, tagging spans as persons (PER) and other categories; (ii) *alias clustering*, so that the multiple surface forms of a character (*Elizabeth*, *Lizzy*, *Miss Bennet*) are grouped into one coreference cluster; and (iii) *coreference resolution*, which attaches anaphoric pronouns (*she*, *he*) to the appropriate cluster. From the resulting annotations we retain a coreference cluster as a *character* only if it is of type PER, contains at least one proper-name mention, and is mentioned at least three times; the canonical label of a character is its most frequent proper-name form. This procedure discards generic nominal mentions (“the man”) while keeping pronominal mentions, which — having been resolved to their referent — count towards the presence of the character in a paragraph.

3.3. Network construction

Each mention is assigned to the paragraph in which it occurs. Let $P_0, P_1, \dots$ be the paragraphs of a novel. We slide a window of $W = 8$ consecutive paragraphs (step one) over the text; for every window we take the set of characters mentioned within it and connect every pair of them. The weight w_{ij} of an edge is the number of windows in which characters i and j co-occur, yielding an undirected weighted graph $G = (V, E)$ where V is the set of characters. The eight-paragraph window is large enough to capture characters that interact across a short scene without being adjacent in a single paragraph, in the spirit of context-window co-occurrence networks [4].

3.4. Network measures

For each network we compute vertex-level centralities: degree k , weighted degree (strength), closeness, eigenvector centrality and betweenness centrality C_B . Betweenness is computed with Brandes’ algorithm [8] on the weighted graph, mapping each weight to a distance $d_{ij} = 1/w_{ij}$ so that frequently co-occurring characters are “closer”. At the global level we report the density, the average degree $\langle k \rangle$, the average clustering coefficient C , the average shortest path length L and diameter (on the giant component), the degree assortativity r , and the modularity Q of the partition returned by greedy modularity maximisation, together with the number of communities. To assess the small-world character [12] we compute $\sigma = (C/C_{\text{rand}})/(L/L_{\text{rand}})$, where C_{rand} and L_{rand} are averaged over Erdős–Rényi graphs with the same number of vertices and edges. The whole pipeline is implemented in Python with NetworkX [14]; Figure 1 summarises it.

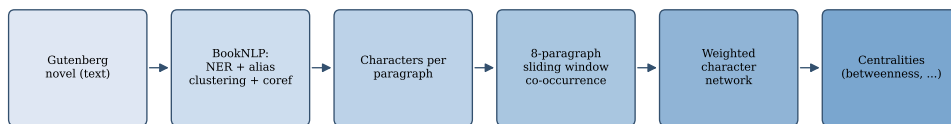


Figure 1: From raw text to analysed network: BookNLP resolves characters, aliases and anaphora; co-occurrence within an eight-paragraph window defines weighted edges; centralities are then computed on the resulting graph.

4. Results and discussion

4.1. Global topology: small-world character networks

Table 2 reports the global measures for the ten networks. All of them combine a high clustering coefficient ($C \approx 0.69$ – 0.79) with a short average path length ($L \approx 1.6$ – 2.1), and the small-world coefficient σ is well above unity in every case (from $\sigma \approx 2.0$ for *Pride and Prejudice* to $\sigma \approx 9.5$ for *The Adventures of Sherlock Holmes*). This confirms, on a new corpus and with a different edge definition, the small-world nature previously reported for named-entity co-occurrence networks [4]. The networks are also consistently disassortative ($r < 0$): high-degree protagonists tend to connect to

low-degree minor characters rather than to one another. Modularity is moderate ($Q \approx 0.08$ – 0.40), with the most episodic works — *Sherlock Holmes*, a collection of self-contained stories, and the seafaring *Moby-Dick* — being split into the largest number of communities. Figure 2 shows three representative networks, with vertex size proportional to betweenness and colours encoding communities.

Table 2: Global network measures: number of vertices N and edges E , average degree $\langle k \rangle$, clustering C , average path length L , assortativity r , modularity Q and small-world coefficient σ .

Novel	N	E	$\langle k \rangle$	C	L	r	Q	σ
Pride and Prejudice	94	1741	37.0	0.79	1.60	-0.31	0.08	2.0
Moby-Dick	245	2716	22.2	0.71	2.08	-0.26	0.29	7.7
Frankenstein	50	297	11.9	0.75	1.94	-0.23	0.35	2.9
A Tale of Two Cities	91	769	16.9	0.69	1.98	-0.20	0.23	3.5
Dracula	112	1019	18.2	0.78	1.87	-0.38	0.11	4.7
Great Expectations	121	1182	19.5	0.74	1.87	-0.31	0.23	4.3
Huckleberry Finn	124	893	14.4	0.71	2.01	-0.19	0.37	6.2
The Adventures of Sherlock Holmes	105	531	10.1	0.78	1.92	-0.23	0.29	9.5
Emma	120	2031	33.9	0.79	1.72	-0.39	0.08	2.8
Jane Eyre	147	1420	19.3	0.74	1.98	-0.24	0.40	5.6

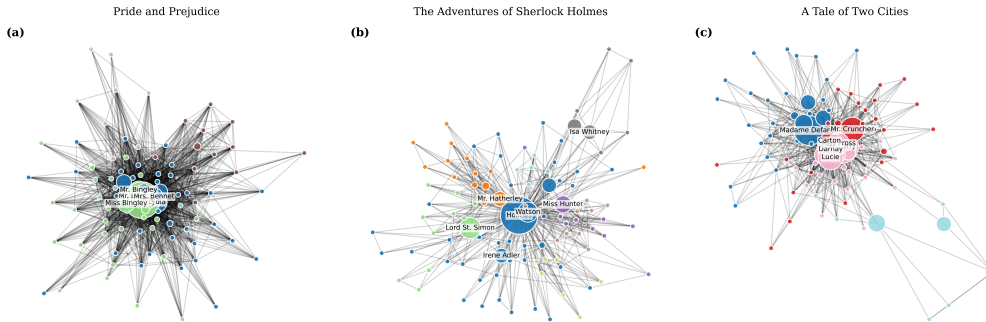


Figure 2: Representative character co-occurrence networks. Vertex size is proportional to betweenness centrality; colours indicate communities obtained by greedy modularity maximisation. (a) *Pride and Prejudice*, (b) *The Adventures of Sherlock Holmes*, (c) *A Tale of Two Cities*.

4.2. The most “between” character

Table 3 and Figure 3 give the central result: the character with the highest betweenness in each novel. The ranking is linguistically sensible and, in several cases, more revealing than a simple frequency count. In *A Tale of Two Cities* the top vertex is *Mr. Lorry*, the banker who literally shuttles between the English and French halves of the plot — a textbook bridge. In *Pride and Prejudice*, *Emma* and *Sherlock Holmes*, betweenness recovers the obvious protagonist (*Elizabeth*, *Emma*, *Holmes*).

A striking pattern emerges when betweenness is compared with degree (Table 3): the two measures disagree precisely in first-person narratives. In *Great Expectations*, *Joe* has the largest betweenness although the narrator *Pip* has the largest degree; in *Jane Eyre*, the narrator *Jane* leads in betweenness while *Mrs. Fairfax* leads in degree. The reason is methodological and interpretable: a first-person narrator is rendered overwhelmingly through the pronoun “I”, which coreference assigns to the narrator but which does not generate co-mentions in the same way explicit naming does; consequently another character often becomes the structural connector. This is visible at the corpus level in Figure 3, where first-person narratives (red) populate both extremes of the betweenness scale.

Table 3: Character with the highest betweenness C_B and the highest degree k in each novel. The two criteria diverge mainly in first-person narratives.

Novel	Top betweenness	C_B	Top degree	k
Pride and Prejudice	Elizabeth	0.876	Elizabeth	89
Moby-Dick	Ahab	0.663	Ahab	165
Frankenstein	Elizabeth	0.550	Elizabeth	36
A Tale of Two Cities	Mr. Lorry	0.695	Mr. Lorry	60
Dracula	Jonathan	0.589	Jonathan	98
Great Expectations	Joe	0.496	Pip	101
Huckleberry Finn	Jim	0.901	Jim	108
The Adventures of Sherlock Holmes	Holmes	0.971	Holmes	102
Emma	Emma	0.861	Emma	118
Jane Eyre	Jane	0.662	Mrs. Fairfax	113

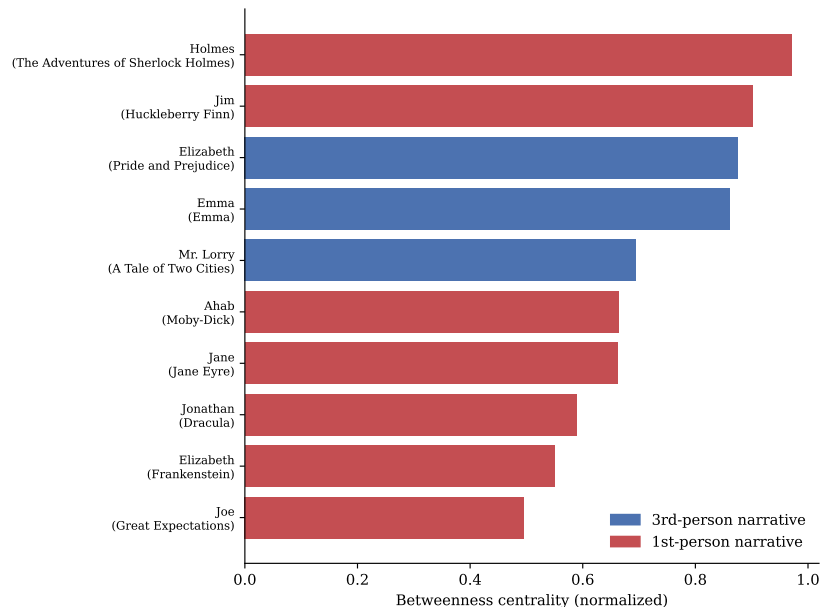


Figure 3: Top betweenness value per novel, coloured by narrative point of view.

4.3. *Betweenness captures a distinct notion of importance*

Pooling all characters across the ten books, Figure 4 shows the Spearman correlations between centrality measures. Degree, weighted degree, closeness and eigenvector centrality are strongly inter-correlated, as expected, whereas betweenness is the measure least aligned with the others. In other words, the most frequent or most connected character is not necessarily the one that mediates information flow through the narrative — exactly the property that makes betweenness valuable for identifying connector characters.

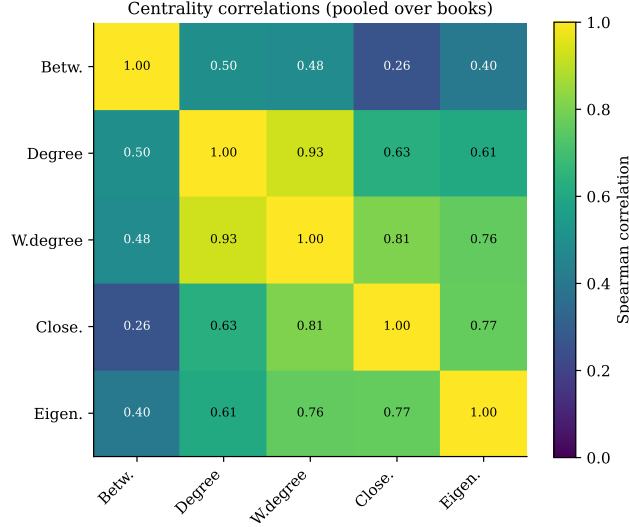


Figure 4: Spearman correlation between centrality measures, pooled over all characters of the ten novels. Betweenness is the least correlated with the remaining measures.

4.4. Narrative point of view and the concentration of betweenness

Figure 5 relates the inequality of the betweenness distribution (measured by its Gini coefficient) to the maximum betweenness, separating first- and third-person novels. Third-person novels with a clear central protagonist (*Emma*, *Pride and Prejudice*) reach the highest peak betweenness, since one vertex dominates the shortest-path structure. First-person novels show a broader range, consistent with the narrator-pronoun effect discussed above. These observations should be read together with the choice of the `small` BookNLP model and the eight-paragraph window; both are stated explicitly to keep the study reproducible, and neither affects the qualitative conclusions.

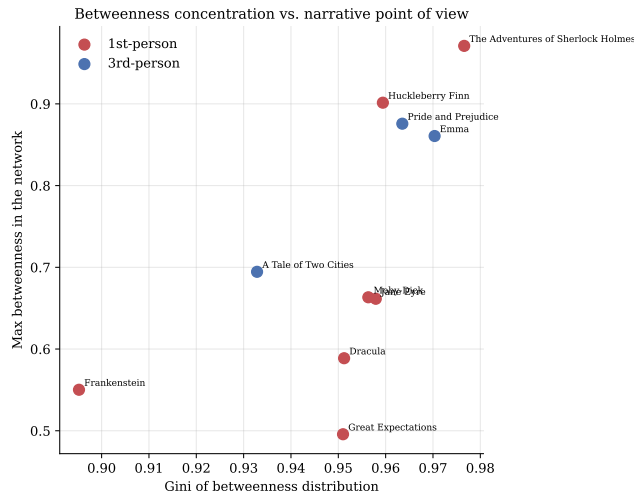


Figure 5: Concentration of betweenness (Gini coefficient) versus the maximum betweenness, separated by narrative point of view.

5. Conclusion

We built weighted character co-occurrence networks for ten classic English novels using a neural literary-NLP pipeline that resolves characters, aliases and anaphora, and connected characters co-occurring within an eight-paragraph window. The networks are robustly small-world and disassortative, and their betweenness rankings identify the *connector* characters of each story, which need not coincide with the most frequent ones — a divergence that is systematic in first-person narratives. Betweenness emerges as the centrality least correlated with the others, underlining its complementary, mediation-based view of narrative importance. Future work includes scaling the analysis with the larger BookNLP model, contrasting co-occurrence with conversational edges [6], varying the context window, and extending the corpus to enable network-based classification of authorship and genre along the lines of [1, 3]. All code, intermediate artefacts and per-book networks are released to ensure full reproducibility.

Acknowledgements

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References

- [1] D. R. Amancio, A complex network approach to stylometry, PLOS ONE 10 (8) (2015) e0136076. doi:10.1371/journal.pone.0136076.
- [2] D. R. Amancio, Probing the topological properties of complex networks modeling short written texts, PLOS ONE 10 (2) (2015) e0118394. doi:10.1371/journal.pone.0118394.
- [3] D. R. Amancio, O. N. Oliveira Jr., L. d. F. Costa, Identification of literary movements using complex networks to represent texts, New Journal of Physics 14 (4) (2012) 043029. doi:10.1088/1367-2630/14/4/043029.
- [4] D. R. Amancio, Network analysis of named entity co-occurrences in written texts, EPL (Europhysics Letters) 114 (5) (2016) 58005. doi:10.1209/0295-5075/114/58005.
- [5] V. Labatut, X. Bost, Extraction and analysis of fictional character networks: A survey, ACM Computing Surveys 52 (5) (2019) 89:1–89:40. doi:10.1145/3344548.
- [6] D. K. Elson, N. Dames, K. R. McKeown, Extracting social networks from literary fiction, in: Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics (ACL), 2010, pp. 138–147.
- [7] L. C. Freeman, A set of measures of centrality based on betweenness, Sociometry 40 (1) (1977) 35–41. doi:10.2307/3033543.
- [8] U. Brandes, A faster algorithm for betweenness centrality, The Journal of Mathematical Sociology 25 (2) (2001) 163–177. doi:10.1080/0022250X.2001.9990249.
- [9] D. Bamman, T. Underwood, N. A. Smith, A bayesian mixed effects model of literary character, in: Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (ACL), 2014, pp. 370–379. doi:10.3115/v1/P14-1035.
- [10] M. Honnibal, I. Montani, spaCy 2: Natural language understanding with Bloom embeddings, convolutional neural networks and incremental parsing, 2017, software available from <https://spacy.io>.

- [11] M. E. J. Newman, *Networks: An Introduction*, Oxford University Press, 2010. doi:10.1093/acprof:oso/9780199206650.001.0001.
- [12] D. J. Watts, S. H. Strogatz, Collective dynamics of ‘small-world’ networks, *Nature* 393 (6684) (1998) 440–442. doi:10.1038/30918.
- [13] Project Gutenberg, Project Gutenberg, <https://www.gutenberg.org>, accessed: 2026-06-28.
- [14] A. A. Hagberg, D. A. Schult, P. J. Swart, Exploring network structure, dynamics, and function using NetworkX, in: *Proceedings of the 7th Python in Science Conference (SciPy)*, 2008, pp. 11–15.

Appendix A. Per-novel networks and character rankings

This appendix collects, for each of the ten novels, the full character co-occurrence network and the ten most central characters ranked by betweenness.

Appendix A.1. Pride and Prejudice (Austen, 1813)

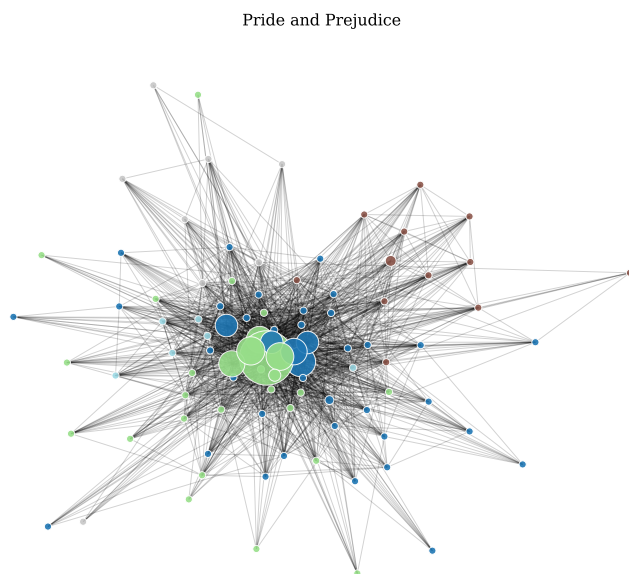


Figure A.6: Character co-occurrence network of *Pride and Prejudice*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Elizabeth	0.876	89	27.15	0.488
Lydia	0.072	83	26.27	0.245
Mr. Darcy	0.058	85	26.64	0.352
Jane	0.053	86	26.63	0.357
Mrs. Bennet	0.044	86	26.26	0.268
Mr. Bingley	0.043	78	26.18	0.252
Miss Bingley	0.042	62	25.60	0.149
Lady Catherine	0.022	74	26.22	0.255
Sir William	0.022	57	23.62	0.077
Mr. Bennet	0.021	75	25.57	0.175

Table A.4: Top-10 characters in *Pride and Prejudice* ranked by betweenness.

Appendix A.2. *Moby-Dick* (Melville, 1851)

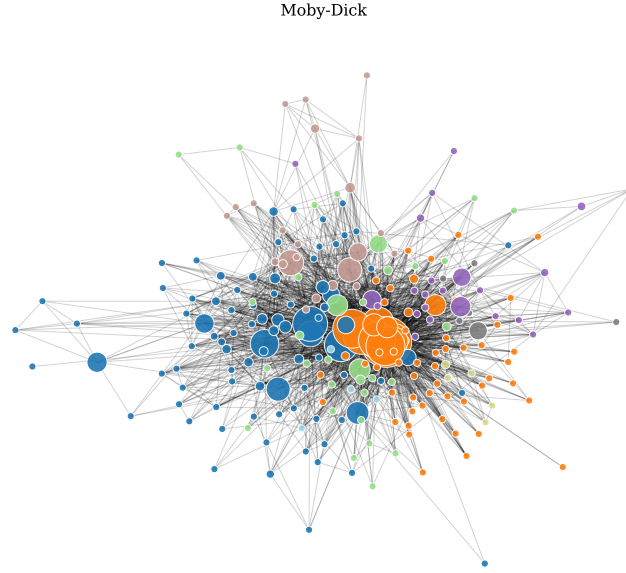


Figure A.7: Character co-occurrence network of *Moby-Dick*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Ahab	0.663	165	10.36	0.490
Stubb (#771)	0.224	141	10.25	0.416
God	0.220	134	10.13	0.195
Queequeg	0.191	125	10.02	0.227
the Pequod	0.172	156	10.22	0.322
Leviathan	0.142	101	8.76	0.040
the Sperm Whale	0.115	90	9.55	0.089
Moby Dick	0.086	88	10.04	0.213
Starbuck	0.060	112	10.23	0.403
Jonah	0.060	65	8.03	0.012

Table A.5: Top-10 characters in *Moby-Dick* ranked by betweenness.

Appendix A.3. *Frankenstein* (Shelley, 1818)

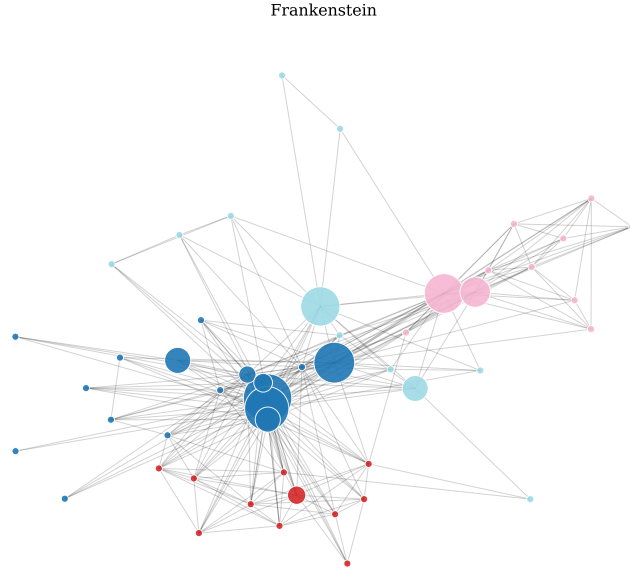


Figure A.8: Character co-occurrence network of *Frankenstein*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Elizabeth	0.550	36	9.66	0.507
Clerval	0.380	36	9.43	0.402
God	0.271	24	9.11	0.257
Felix	0.240	23	6.79	0.050
Frankenstein (#151)	0.233	17	8.39	0.096
Safie	0.079	18	6.32	0.035
Mr. Kirwin	0.041	8	7.34	0.077
Margaret	0.040	11	5.96	0.039
Victor	0.035	26	9.03	0.352
William	0.009	26	8.96	0.366

Table A.6: Top-10 characters in *Frankenstein* ranked by betweenness.

Appendix A.4. *A Tale of Two Cities* (Dickens, 1859)

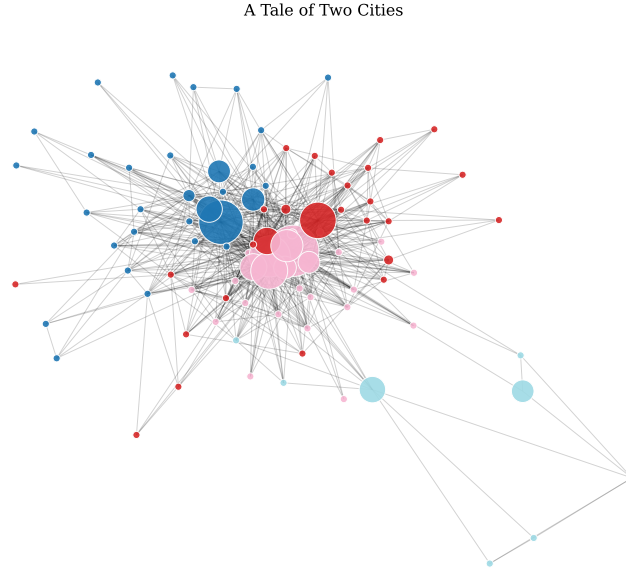


Figure A.9: Character co-occurrence network of *A Tale of Two Cities*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Mr. Lorry	0.695	60	15.11	0.468
Madame Defarge	0.382	48	14.45	0.215
Darnay	0.282	55	14.78	0.404
Lucie	0.179	47	14.60	0.356
Mr. Cruncher	0.171	46	13.62	0.126
Miss Pross	0.100	41	14.48	0.278
Carton	0.050	46	13.74	0.237
Doctor Manette	0.047	47	14.48	0.328
Death	0.044	11	5.43	0.007
Saint Antoine	0.043	36	12.96	0.062

Table A.7: Top-10 characters in *A Tale of Two Cities* ranked by betweenness.

Appendix A.5. *Dracula* (Stoker, 1897)

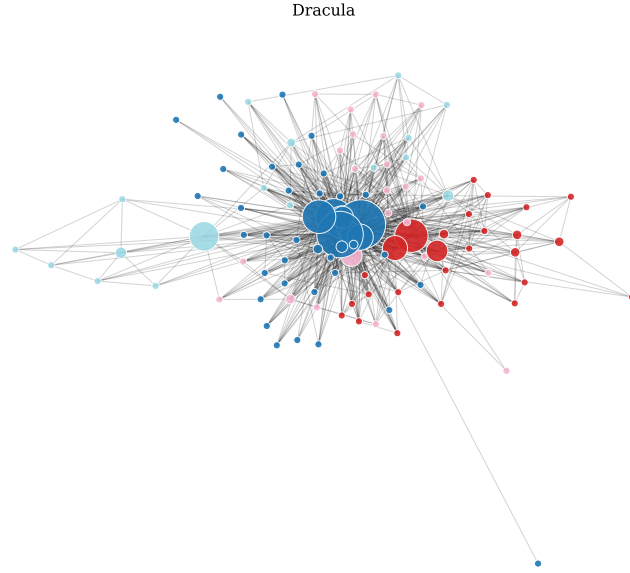


Figure A.10: Character co-occurrence network of *Dracula*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Jonathan	0.589	98	12.92	0.440
Van Helsing (#335)	0.461	86	12.89	0.461
Lucy	0.162	76	12.78	0.361
Renfield	0.119	50	12.11	0.123
Count Dracula	0.118	46	11.47	0.047
Dr. Seward	0.116	88	12.77	0.347
Lor	0.069	21	6.97	0.007
Lord Godalming	0.042	63	12.60	0.247
Galatz	0.035	35	10.95	0.048
Arthur	0.019	69	12.63	0.300

Table A.8: Top-10 characters in *Dracula* ranked by betweenness.

Appendix A.6. *Great Expectations* (Dickens, 1861)



Figure A.11: Character co-occurrence network of *Great Expectations*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Joe	0.496	87	18.77	0.409
Pip	0.456	101	18.88	0.472
Herbert	0.449	84	18.49	0.258
Miss Havisham	0.228	74	18.70	0.427
Wemmick	0.139	59	18.21	0.191
Mr. Jaggers	0.098	63	18.20	0.239
Estella	0.066	70	18.21	0.325
Mr. Pumblechook	0.049	48	17.83	0.176
Sarah Pocket	0.032	28	16.40	0.076
Mr. Pocket	0.027	35	15.45	0.070

Table A.9: Top-10 characters in *Great Expectations* ranked by betweenness.

Appendix A.7. *Huckleberry Finn* (Twain, 1884)

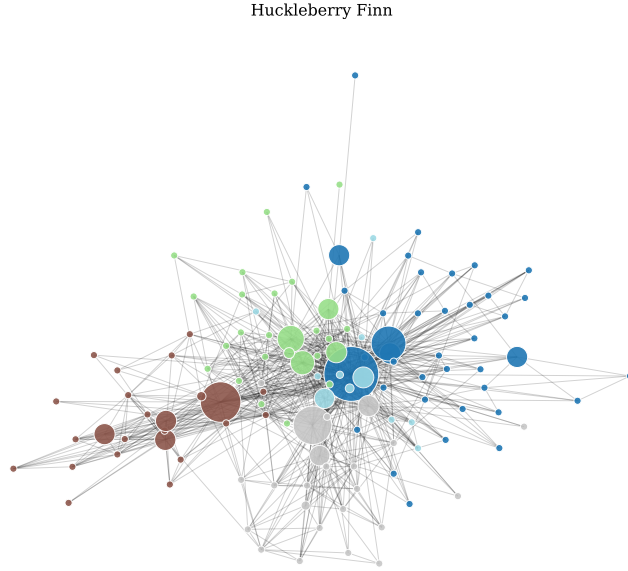


Figure A.12: Character co-occurrence network of *Huckleberry Finn*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Jim	0.901	108	13.02	0.615
Mary Jane	0.263	49	10.57	0.088
Buck	0.211	46	11.07	0.109
Aunt Sally	0.140	44	12.43	0.358
Bill	0.048	35	9.58	0.075
Pap	0.029	42	11.52	0.161
Peter	0.017	26	8.92	0.023
Harvey	0.016	17	9.05	0.021
Dey	0.016	29	10.70	0.089
Sherburn	0.016	20	6.48	0.041

Table A.10: Top-10 characters in *Huckleberry Finn* ranked by betweenness.

Appendix A.8. *The Adventures of Sherlock Holmes* (Doyle, 1892)

The Adventures of Sherlock Holmes

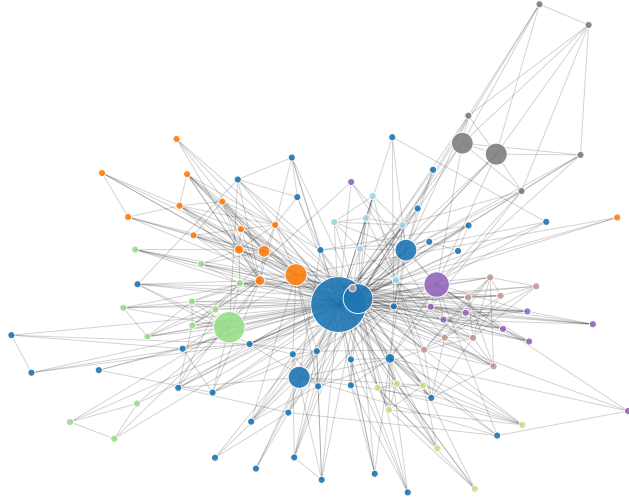


Figure A.13: Character co-occurrence network of *The Adventures of Sherlock Holmes*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Holmes	0.971	102	18.97	0.655
Lord St. Simon	0.093	15	16.62	0.132
Watson	0.073	76	18.34	0.487
Miss Hunter	0.037	18	16.47	0.131
Isa Whitney	0.019	10	8.30	0.019
Mr. Hatherley	0.019	17	14.68	0.058
Irene Adler	0.019	12	15.62	0.081
John Openshaw	0.018	20	14.89	0.084
Kate	0.018	9	8.84	0.021
young McCarthy	0.001	16	16.43	0.134

Table A.11: Top-10 characters in *The Adventures of Sherlock Holmes* ranked by betweenness.

Appendix A.9. *Emma* (Austen, 1815)

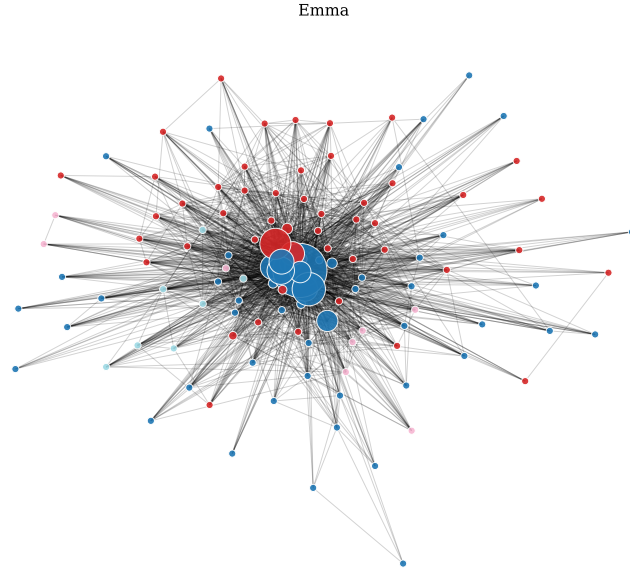


Figure A.14: Character co-occurrence network of *Emma*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Emma	0.861	118	20.05	0.479
Harriet	0.116	103	19.78	0.307
Mrs. Elton	0.087	78	19.32	0.153
Mrs. Weston	0.051	95	19.71	0.299
Mr. Knightley (#281)	0.034	100	19.76	0.320
Mr. Weston	0.034	88	19.54	0.236
Mr. Elton	0.033	92	19.55	0.212
Miss Fairfax	0.028	93	19.66	0.250
Miss Woodhouse	0.019	101	19.62	0.261
Mr. Woodhouse	0.017	103	19.50	0.220

Table A.12: Top-10 characters in *Emma* ranked by betweenness.

Appendix A.10. *Jane Eyre* (Brontë, 1847)

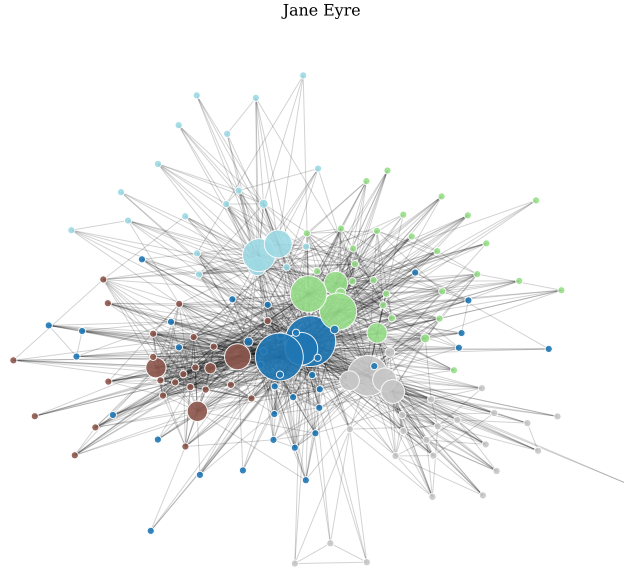


Figure A.15: Character co-occurrence network of *Jane Eyre*. Node size is proportional to betweenness and colours denote communities (greedy modularity); the most central characters are named in the accompanying table.

Character	C_B	Degree	Closeness	Eigenvector
Jane	0.662	103	17.52	0.460
Mr. Rochester	0.523	101	17.42	0.498
St. John	0.265	66	16.81	0.252
Bessie	0.181	72	16.72	0.195
Mrs. Reed	0.171	65	16.66	0.139
Miss Temple	0.111	31	14.99	0.053
Mrs. Fairfax	0.107	113	17.30	0.453
Helen	0.054	26	14.85	0.039
Miss Ingram	0.040	50	16.24	0.131
Mary	0.027	40	15.99	0.125

Table A.13: Top-10 characters in *Jane Eyre* ranked by betweenness.